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November 15, 1971
SUPERSEDING
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April 24, 1965

FEDERAL SPECIFICATION

FILM, DIAZOTYPE, SENSITIZED (DIRECT IMAGE MICROFORMS)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for positive working, direct image ammonia developed diazotype films intended for use in microreproduction systems. Where applicable, films include those in rolls, sheets and mounted in EAM copy cards.

1.2 Classification. Diazotype filsm covered by this specification shall be classified using the designators specified in table I, and meeting the requirements put forth in table II. The film product's nominal value of the appropriate characteristic will fall within one of the designator groups. The film product supplied by the manufacturer shall not vary more than a fixed percentage of the manufacturer's nominal stated value, as shown on table I. While this designator system includes every conceivable combination of sensitized product, many of these products are not feasible in the diazo process at this time.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-B-640 - Box, Fiberboard, Corrugated, Triple-Wall.
PPP-T-60 - Tape, Pressure Sensitive, Adhesive, Waterproof for packaging.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

FSC 6750

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Military Specifications:

MIL-L-10547 - Liners, Case, Waterproof.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 - Marking for Shipment and Storage.

MIL-STD-147 - Palletized and Containerized Unit Loads.

MS-26565 - Spool, Photographic Film.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring acitivity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

American National Standards Institute (ANSI) Standards:

PH1.13 - Cores for Photographic Film Rolls (Plastic, Wood, or Metal), Dimensions for.

PH1.16 - Graphic Arts Sheet Film (Inch and Centimeter Sizes), Dimensions for.

PH1.20 - 70mm Unperforated and Perforated Film for Camera Other than Motion-Picture Cameras, Dimensions for.

PH1.25 - Safety Photographic Film, Specifications for.

PH1.29 - Curl of Photographic Film, Methods for Determining the.

PH1.32 - Determining the Dimensional Change Characteristics of Photographic Films and Papers, Method for.

PH1.33 - 16mm 100-Foot Film Spools for Recording Instruments, Microfilm and Still Picture Cameras, Dimensions for.

PH1.34 - 16mm 200-Foot Film Spools for Recording Instruments, Microfilm and Still Picture Cameras, Dimensions for.

PH1.35 - 35mm 100-Foot Film Spools for Recording Instruments, Microfilm and Still Picture Cameras, Dimensions for.

PH2.2 - Sensitometry of Photographic Papers.

PH2.19 - Diffuse Transmission Density.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, N.Y. 10018.)

Clearing House for Federal Scientific and Technical Information:

Microfiche Standards, Third Edition, April 1968.

(Copies of the Microfiche Standard are available from the Clearning House for Federal Scientific and Technical Information, Sils Building, 5285 Port Royal Road, Springfield, Virginia 22171.

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National Motor Freight Traffic Association, Incorporated, Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, D. C. 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 202 Union Station, 516 W. Jackson Blvd., Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Materials. The diazotype film procured in accordance with this specification shall meet all the requirements specified herein. Any material which exhibits coating or physical defects not meeting the specification shall be replaced by the manufacturer at no additional cost to the Government.

3.2 Manufacturing defects. When examined for the physical defects listed in 4.4.2.1.1, as specified in 4.3.1, the film shall be free from defects of a physical nature that will impair the end use of the product for duplicating microfilms.

3.3 Expiration date. Unless otherwise specified (see 6.2), the expiration date of the diazotype microfilm shall be not less than 90 days from the date of shipment.

3.4 Classification designators, (table I).

3.4.1 Film base. The film base shall be an optically, homogenous, transparent medium. The film base shall comply with only the burning requirements of ANSI PH1.25, and as defined therein for "Safety Photographic Film."

3.4.2 Cellulose ester base. Cellulose ester base shall conform to ANSI PH1.25.

3.4.3 Dimensional stability. The film shall meet the dimensional stability requirements specified in table II, when tested in accordance with 4.3.3.

3.4.4 Total product thickness. The nominal total product thickness of Format 1, 2 and 3 film (table II) shall conform to the designator in table I when tested immediately after slitting in accordance with 4.3.4. Thickness variations from the manufacturer's nominal shall be within ± 10 percent.

3.4.5 Development type high temperature - ambient pressure. This diazo film is designed to develop satisfactorily in equipment utilizing high temperature, 120° to 220°F., and ambient pressure as development conditions.

3.4.6 Development type ambient temperature - high pressure. This diazotype film is designed to develop satisfactorily in equipment utilizing ambient temperature and high pressure, above 30 lbs. per square inch as development conditions.

3.4.7 Film usage. The film usage shall conform to the designator in table I; a "print" film designed for viewing, blowbacks, or reprinting; and a sepia "intermediate" used solely as a duplicate master film.

3.4.8 ARMM speed rating. The nominal ARMM speed designator of the film shall conform to the values in table I when tested in accordance with 4.3.5. Speed variations from the manufacturer's nominal shall be within ± 20 percent.

3.4.9 Bar gamma. The nominal bar gamma designator of film specified herein shall conform to the values in table I when tested as specified in 4.3.6. Variations from the manufacturer's nominal shall be within ± 10 percent.

3.4.10 Bar gamma ratio. The bar gamma ratio designator of the film specified herein shall meet the values shown in table I when tested in accordance with 4.3.7. Bar gamma ratio variations from the manufacturer's nominal shall be within ± 10 percent.

3.4.11 Maximum density. The maximum density designator of films specified herein shall be in accordance with the values in table I when tested as specified in 4.3.8. Maximum density variations from the manufacturer's nominal shall be within ± 10 percent.

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TABLE I.

DIAZOTYPE FILM CLASSIFICATION DESIGNATORS

Support Type	Total* Product Thickness (inches)	Development Type	Film Usage	ARMM** Speed	Bar* Gamma	Bar* Gamma Ratio	Maximum* Density
1. Cellulose - Ester	1. 0.0012 - .0019	1. High temperature - ambient pressure	1. Print	1. 1-10	1. 0.50 - 0.90	1. under 1.00	1. under 1.00
2. Polyester	2. .0020 - .0026	2. Ambient temperture - high pressure	2. Intermediate (Sepia)	2. 11-20	2. .91-1.40	2. 1.00-1.50	2. 1.00-1.50
	3. .0027 - .0036			3. 21-30	3. 1.41-1.80	3. 1.51-2.00	3. over 1.50
	4. .0037 - .0046			4. 31-40	4. over 1.80	4. over 2.00	
	5. .0047 - .0063			5. 41-50			
	6. .0064 - .0087			6. 51-60			
	7. .0088 - .0120			7. 61-70			
	8. Other			8. over 70			
1	1 5	1	1	5	3	3	3 ***

* The nominal value of each of the film product characteristics marked with asterisk is to fall within the appropriate designater, with the actual product supplied to vary within ± 10 percent of the manufacturer's nominal value.

** The nominal value of each of the film product characteristics marked with asterisk is to fall within the appropriate designator, with the actual product supplied to vary within ± 20 percent of the manufacturer's nominal value.

*** Sample Designator. A diazo microfilm on cellulose ester base of 0.0047 - 0.0063 inch thickness, designed for duplicating machines utilizing high temperature and ambient pressure conditions for development, to be used as a print film with an ARMM speed between 41 and 50, a Bar Gamma of 1.41 to 1.80, a Bar Gamma Ratio of 1.51 to 2.00, and a visual density of over 1.50.

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3.5 Product characteristics (table II).

3.5.1 Format. Three formats as supplied by the manufacturer are specified in table II.

3.5.2 Width format 1 and 2. Format 1 and 2 films shall conform to the width in table II when tested immediately after slitting in accordance with 4.3.9.

3.5.3 Size format 3 . Format 3 film shall conform to the size specifications of page 1 of PB 167/630, Microfiche Standards, Third Edition, April 1968, or its latest revision.

3.5.4 Splices. Format 1 and 2 film supplied in 1000 and 2000 foot lengths may contain one splice, provided that not more than 10 percent of such spliced rolls are included in any one shipment.

3.5.5 Spools and cores. Unless otherwise specified by the ordering activity (see 6.2), spools for 16mm by 100 feet shall be in accordance with ANSI PH1.33. Spools for 16 mm by 200 feet shall be in accordance with ANSI PH1.34. Spools for 35mm by 100 feet shall be in accordance with ANSI PH1.35. Spools for 70mm shall be in accordance with MS-26565. Cores shall be in accordance with ANSI PH1.13. Commercial spools or cores not covered by the above specified ANSI and MS Standards may be supplied when the contractor has certified that the performance of the commercial spools or cores are equal to or better than the standard spools or cores (see 6.2).

3.5.6 Roll lengths. Format 1 and 2 film shall conform to the roll length in table II.

3.5.7 Number of sheets per package. Format 3 film shall conform to the number of sheets per package in table II.

TABLE II. Characteristics - designators

A. Format (3.5.1)

1. Rolls - unperforated
2. Rolls - perforated
3. Sheets

B. Width* (3.5.2 and 3.5.3)

16mm - 0.626 ± 0.002 inches

35mm - 1.377 ± 0.002 inches

70mm - 2.754 ± 0.005 inches

Tab Card - 3.245 ± 0.005 inches

105mm - 4.129 ± 0.005 inches

3-1/4 inch - 3.245 ± 0.005 inches

Microfiche - 105 x 148.75mm, +0 -0.75mm

3.250 x 7.375 inches (+0 -0.030 inches)

Unless otherwise specified
*Measured at 72°F. ± 2°F. and 50 percent RH ± 2 percent RH.
Other sizes cut to ANSI Standard PH1.16.

C. Roll length** (3.5.6)

100, 200, 300, 400, 500, 1000, 2000 ft.

**A combined integral film leader and trailer not less than 4ft. in length shall be included on each roll.

D. Number of sheets per package (3.5.7) 100, 500

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TABLE II. (CON.)

E. Minimum Density (3.5.9) 0.12 maximum (see 4.3.10).

F. Actinic Transmission (3.5.10) 30 percent minimum (see 4.3.11).

G. Resolving Power (3.5.11) 150 lines per millimeter minimum (see 4.3.12).

H. Accelerated Aging Characteristics (see 3.5.13 and 4.3.13).

	Unprocessed Film		Processed Film	
	Minimum Density Change	Maximum Density Change	Minimum Density Change	Maximum Density Change
Light Aged Samples	-	-	0.10 density maximum (see 4.3.13.1).	35 percent maximum (see 4.3.13.2).
Heat Aged Samples	0.10 density maximum (see 4.3.13.3)	20 percent maximum (see 4.3.13.4)	0.05 density maximum (see 4.3.13.5).	15 percent maximum (see 4.3.13.6).

I. Development (3.5.13) 90 percent in one pass (see 4.3.14).

J. Curl - Format 1 and 2 (3.5.14.1) 60 maximum

K. Curl - Format 3 (3.5.14.2) 0.25 inch maximum

L. Dimensional Stability of Sensitized Films (see 4.3.3).

	Humidity Coefficient % per 1% RH	Thermal Coefficient % per 1°F.	Permanent Size Change Humidity Cycle Percent	Permanent Size Change Temperature Cycle Percent
Polyester	0.0030 max.	0.0020 max.	± 0.10	± 0.10
Acetate	0.020 max.	0.010 max.	± 0.30	± 0.25

3.5.8 Perforations. Perforations when specified (see 6.2), shall be type II perforations in accordance with ANSI PH1.20.

3.5.9 Minimum density. The minimum density of films specified herein shall be according to the values in table II when tested as specified in 4.3.10.

3.5.10 Actinic transmittance. The actinic transmission shall conform to the value in table II when tested as specified in 4.3.11.

3.5.11 Resolving power. Resolving power shall conform to the value in table II when tested, as specified in 4.3.12.

3.5.12 Accelerated aging characteristics.

3.5.12.1 Change of minimum density of light aged, processed film samples. When tested, as specified in 4.3.13.1, the change in minimum density shall comply to table II.

3.5.12.2 Change of maximum density of light aged, processed film samples. When tested, as specified in 4.3.13.2, the change of maximum density shall comply to table II.

3.5.12.3 Change of minimum density of heat aged, unprocessed film samples. When tested, as specified in 4.3.13.3, change of minimum density shall comply to table II.

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3.5.12.4 Change of maximum density of heat aged, unprocessed film samples. When tested, as specified in 4.3.13.4, change of maximum density shall comply to table II.

3.5.12.5 Change of minimum density of heat aged, processed film samples. When tested, as specified in 4.3.13.5, the change of minimum density shall comply to table II.

3.5.12.6 Change of maximum density of heat aged, processed film samples. When tested, as specified in 4.3.13.6, the change of maximum density shall comply to table II.

3.5.13 Developing rate. Developing rate for the film specified herein shall comply to table II when obtained in one pass through the developing section as specified in accordance with 4.3.14.

3.5.14 Curl.

3.5.14.1 Format 1 and 2. The curl of processed and unprocessed format 1 and 2 film shall comply to table II when tested in accordance to Method B of ANSI PH1.29. Processing of samples before testing shall be carried out on equipment recommended by the manufacturer.

3.5.14.2 Format 3. The curl of processed and unprocessed Format 3 film when supplied in sheet form shall be measured as specified in and conform to paragraph C. 3 of PB 167'630 Microfiche Standards. Third Edition, April 1968, or its latest revision. Processing of samples before measurement shall be carried out on equipment recommended by the manufacturer.

4. SAMPLING AND INSPECTION

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for inspection and acceptance. Unless otherwise specified (see 6.2), sampling for inspection and acceptance shall be performed in accordance with MIL-STD-105. For purposes of sampling an inspection lot for examination and testing shall consist of all materials of the same classification submitted for delivery at one time.

4.2.1 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for ensuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications. In the event of conflict, this specification shall govern.

4.2.2 Inspection of the end item.

4.2.2.1 Examination of the end item. Examination of the end item shall be made in accordance with the classification of defects, inspection levels and acceptable quality levels (AQLs) set forth in 4.2.2.1.5. Not more than 3 rolls of roll film or not more than one package of sheet film shall be drawn from any one shipping container for each lot of film offered for acceptance for visual and dimensional characteristics. The lot size for purposes of determining sample size in accordance with MIL-STD-105 shall be expressed in units of rolls or packages of sheets for examination under 4.2.2.1.1, 4.2.2.1.2 and 4.2.2.1.3 and in units of shipping containers for examination on 4.2.2.1.4. Examinations included in 4.2.2.1.1, 4.2.2.1.2, and 4.2.2.1.3 may be performed at any point immediately before, during or after the spooling or packaging operation.

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4.2.2.1.1 Examination of the end item for defects in appearance and workmanship. The sample unit for this examination shall be one linear foot or one sheet of sensitized film as applicable. Not more than three sample units shall be drawn from any one roll or package.

Examine	Defects
Appearance and workmanship	Film not completely or evenly coated. Emulsion surface shows scuffs, scratches, foreign matter or abrasions.
	Abnormal curl.
	Perforation holes not clearly punched.
	Holes, pinholes, tears, cuts, cracks, finger marks, streaks, spots, or sharp creases.

4.2.2.1.2 Examination of the end item for dimensional defects. The sample unit for this examination shall be one linear foot or one sheet of film as applicable.

Examine	Defects
Width.	Not specified width.
Perforation.	Film perforations do not conform to applicable ANSI Standard.

4.2.2.1.3 Examination of the end item for defects in construction, length and blocking. The sample unit for this examination shall be one roll of film or one package of sheets.

Examine	Defects
Roll	Spool not specified type, damaged.
	Core not specified type, damaged.
	Film unevenly or improperly wound less than specified number of feet.
	Splices poorly made.
	Roll not continuous.
	Roll contains more than allowable splices.
	Sticky winding.
Sheet	Less than specified number of film.
	Sheets of film stick together.

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4.2.2.1.4 Examination of preparation for delivery. An examination will be made to determine that marking, materials, workmanship, contents, and weights comply with the requirements of section 5. Defects will be scored in accordance with the classification below. The sample unit shall be one shipping container, fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects.

Examine	Defects
Packaging and preservation.	Not as specified, not in accordance with contract requirements.
	Rolls or sheets not unit wrapped and packaged as specified.
	Unit packages not uniform.
	Containers of improper construction.
	Not packaged as specified.
Packing.	Not packed as specified.
	Packing material not as specified; closures not accomplished by specified or required methods or materials.
	Number of rolls or packages per carton less than indicated or specified.
Markings (Interior and Exterior)	Marking not in accordance with contract specifications.
	Labels when used not properly attached.
	Markings illegible, incorrect, incomplete, omitted, or not in accordance with contract requirements.
Weight (Interior and exterior)	Gross weight exceeds contract requirements.

4.2.2.1.5 Inspection levels and AQLs for examinations. The inspection levels, for determining the sample size and the AQLs expressed in defects per 100 units, shall be as follows:

Examination	Inspection level	AQL
4.2.2.1.1	1	1.0
4.2.2.1.2	S-3	2.5
4.2.2.1.3	S-2	2.5
4.2.2.1.4	S-4	4.0

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4.2.3 Testing of the end item. Testing of the end item shall be conducted in accordance with table III for the characteristics as indicated therein on each lot submitted for acceptance. The sample unit, as referenced herein is defined as the amount of material required to perform all tests at one time. The sample unit for all testing shall consist of one roll or one package of film. Three sample units, each drawn from a different shipping container, shall be tested with no evidence of failure to meet sample unit or average requirements, as applicable.

TABLE III. Instructions for testing (Sample unit)

Characteristic	Specification Reference		Requirements applicable to Individual Unit	No. Determinations Per Sample Unit	Results Reported as	
	Requirement	Test Method			Pass Fail	Numerically to nearest
Resolving Power	3.5.11	4.3.12	X	Avg. of 5	X	-------
Sensitometric Characteristics	3.4.8	4.3.5	X	Avg. of 2	--	1 unit
	3.4.9	4.3.6	X	Avg. of 2	--	0.01 unit
	3.4.10	4.3.7	X	Avg. of 2	--	.01 unit
	3.4.11	4.3.8	X	Avg. of 2	--	.01 unit
	3.5.9	4.3.10	X	Avg. of 2	X	.01 unit
	3.5.10	4.3.11	X	Avg. of 2	X	1 unit
	3.5.13	4.3.14	X	Avg. of 2	X	---------
Dimensional Stability	3.4.3	4.3.3				
Humidity Coefficient		4.3.3.1	X	Avg. of 5	X	
Thermal Coefficient		4.4.3.2	X	Avg. of 5	X	
Humidity Change		4.3.3.3	X	Avg. of 5	X	
Thermal Change		4.3.3.4	X	Avg. of 5	X	
Width and Size	3.5.2 3.5.3	4.3.9	X	Avg. of 5	X	
Roll length	3.5.6	-------	X	Avg. of 2	X	
Number of Sheets	3.5.7	-------	X	Avg. of 2	X	
Accelerated Aging	3.5.12.1	4.3.13.1		Avg. of 2	X	
	3.5.12.2	4.3.13.2		Avg. of 2	X	
	3.5.12.3	4.3.13.3		Avg. of 2	X	
	3.5.12.4	4.3.13.4		Avg. of 2	X	
	3.5.12.5	4.3.13.5		Avg. of 2	X	
	3.5.12.6	4.3.13.6		Avg. of 2	X	
Curl	3.5.14.1	--------		Avg. of 5	X	
	3.5.14.2	--------		Avg. of 5	X	

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4.3 Test methods.

4.3.1 Manufacturing defects. A sample of film shall be selected at random. For format 1 and 2 material it shall be not less than one linear yard in length and the full width of the material under test. For format 3 material it shall consist of two sheets. The sample shall be cut in half. One-half of the sample shall be put aside, protected from exposure to light. The unprotected half shall be examined in visible light to determine conformance with requirements specified in 3.2. The protected half of the sample shall be partially exposed to produce approximately one-half the maximum density, and then be completely developed on applicable equipment. It shall be visually examined for compliance with requirements specified in 3.2.

4.3.2 Density. All densities specified herein refer to diffuse visual transmission densities Type VI-b in accordance with ANSI PH2.19, and shall be measured on a densitometer conforming to that specification. Note: Densitometers utilizing a photo tube with a S4 response generally require a Wratten 106 filter plus an IR Corning 4-94 infrared rejection filter, and a 2mm aperture.

4.3.3 Dimensional stability.

4.3.3.1 Humidity coefficient of expansion. The humidity coefficient of expansion shall be determined on unprocessed film in accordance with ANSI PH1.32, Section 6. Measurements should be made by Method A - pin gage, although Method B or C may be used.

4.3.3.2 Thermal coefficient of expansion. The thermal coefficient of expansion shall be determined on unprocessed film in accordance with ANSI PH1.32, Section 7. Measurements should be made by Method A - pin gage, although Method B or C may be used.

4.3.3.3 Permanent change (humidity cycle). Precondition unprocessed samples at 70° ± 3°F. and 50 ± 3 percent relative humidity for 16 hours and measure. (Method A, ANSI PH1.32.) Expose the samples to an atmosphere of 70° ± 3°F. and 90 ± 3 percent relative humidity for 24 hours, then recondition under the original conditions for 16 hours and remeasure. The difference between the original measure and measure after treatment shall be recorded as the change due to humidity. Determine the percent change by dividing the original measure into the change, and multiplying by 100.

4.3.3.4 Permanent change (temperature cycle). Precondition unprocessed samples at 70° ± 3°F. and 50 ± 3 percent relative humidity for 16 hours and measure. (Method A, ANSI PH1.32.) Place the samples in a circulating air oven maintained at 180° ± 3°F. for one hour. Recondition for 16 hours at the original conditions and remeasure. The difference between the original measure and the measure after treatment shall be recorded as the change due to temperature. Determine the percent change by dividing the original measure into the change, and multiplying by 100.

4.3.4 Thickness. The test for compliance to total product thickness designator shall be on an unprocessed piece of film measured immediately after slitting. A micrometer caliper which is capable of measuring thickness accurately to 0.0001 inch is to be used.

4.3.5 Speed rating; primary method. The speed ratings of a diazotype film specified herein shall be determined as follows, using an association of reproduction materials, manufacturers (ARMM) sensitometer, or equivalent and a densitometer, as specified in 3.4.2.

After the requisite warm-up and calibration of the sensitometer, exposures are made on the test films for various lengths of time until a density of 0.50 to 0.60 is obtained. Care must be taken to insure complete development on the equipment recommended by the film manufacturer. Using the exposure time that produced the 0.50 to 0.60 density as a starting point, expose a series of test films increasing each exposure time by a factor of 10 percent over the previous exposure time until a density of 0.20 is reached. Measure the density of the exposed and developed samples. Plot the points on coordinate paper using the times of exposure in seconds for the absissa (X axis) and the corresponding densities as the ordinate (Y axis) and draw a smooth curve of best fit through the points. From the curve, accurately determine the exposure time in seconds to expose the sample to density of 0.35. From this time and the sensitometer constant, the energy to expose the sample to a density of 0.35 gross is calculated in units of watt seconds per square centimeter. The reciprocal of this value is then calculated. The ARMM speed rating is then found on Graph I. (see page 17).

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4.3.5.1 Speed rating; alternate method. The following alternate method of determining the speed rating of a diazotype material can also be used by comparing its printing speed to that of a standard diazotype material of a known speed. Such a known standard is the ARMM (Association of Reproduction Materials Manufacturer's) diazo microfilm speed standard film, available from ARMM, 901 N. Washington Street, Alexandria, Virginia 22313. Under this method the ARMM diazo microfilm speed standard film is exposed directly to the lamp (ultra violet) on a commercial diazo microfilm duplicator for a period of time to produce a visual density of 0.35 (gross). A master is not to be incorporated. A tachometer is to be used to accurately determine the speed in inches or feet per minute on the commercial diazo microfilm duplicator which utilized a moving exposure. A stop watch is to be used to accurately determine the exposure time in seconds on a piece of commerical diazo microfilm duplicator which utilized a stationary exposure. Development is to be carried out in the commercial duplicator. This procedure is repeated with the diazo type material under test. The speed of the test material is calculated utilizing the following formula:

$$\text{ARMM Speed of Test Film} = \frac{(36^*) \times (\text{exposure time in seconds on ARMM Standard Film to produce 0.35 density})^{**}}{(\text{Exposure time in seconds on test film to produce 0.35 density})^{**}}$$

*36 is the speed of the ARMM Diazo Microfilm Speed Standard. If the standard you are using has a different speed rating printed on the package, use that number in place of the 36 in the formula.

**Exposure speed in inches or feet per minute can be used with the following formula:

$$\text{ARMM Speed} = \frac{(36^*) \times (\text{Exposure Speed in inches or feet per minute on test film to produce 0.35 density})}{(\text{Exposure speed in inches or feet per minute on ARMM standard film to produce 0.35 density})}$$

4.3.6 Bar gamma. A neutral density silver step tablet composed of 21 graduated steps covering a density range of from approximately 0.1 to 3.0 in increments of 0.15 shall be used as the master. A Kodak #3 step tablet or equal meets these criteria. Using such a step tablet as the master, expose the material under test on a non-reflective black backing, in a commercial diazo film processor recommended by the film manufacturer and develop it as recommended. Exposure and development shall be such that the reproduced minimum density area of the step tablet is approximately 0.02 above burnout density. After development, heat the film in an air circulating oven for five minutes at $122^{\circ} \pm 2^{\circ}F.$ to insure complete removal of the ammonia vapor. Measure the visual density of each step in both the master and test film. Using equal linear scales, plot the density of each step of the silver master along with abscissa (X axis) versus the density of the corresponding step in the print along the ordinate (Y axis). Draw a smooth curve of best fit through these plotted points. Determine the bar-gamma of this first generation diazotype print using the method in ANSI PH2.2.

4.3.7 Gamma ratio. Using the diazotype print from 4.3.6 as the master, repeat the test specified in 4.3.6. Plot the density of each step of the second generation diazotype print against the density of the corresponding step of the neutral silver density step tablet. Determine the bar-gamma of the second diazo generation. Divide the bar-gamma of the second diazotype generation by the bar-gamma of the first diazotype generation to obtain the gamma ratio.

4.3.8 Maximum density. Completely develop without exposure, two strips of material under test not less than 12 inches each in length for format 1 and 2 film, or 2 sheets of format 3 film. An equivalent film area for other formats is to be used. Use a commercial diazo film processor as recommended by the film manufacturer. Heat the developed samples in a forced air circulating oven for five minutes at $122^{\circ} \pm 2^{\circ}F.$ to allow retained ammonia vapors to escape. Using a densitometer conforming to 4.3.2 check the zero and sensitivity settings of the densitometer with the appropriate working standards. For format 1 and 2 film take five density readings on the image side of each strip at two inch intervals. For format 3 film take five readings on the image side of each sheet, one in the center and one on each diagonal half-way to the edge. Determine and report the average of the readings as maximum transmitted density.

4.3.9 Width. For format 1 and 2 film a commercial optical comparator shall be used for measurements of compliance to width requirements. The precision of the comparator used shall be such that measurements of a reference standard are repeatable within $\pm$ 0.0001 inch.

4.3.10 Minimum density. On a commercial diazotype film processor recommended by the film manufacturers, expose two strips of the film under test for format 1 and 2 film, not less than 12 inches each in length and the full width of the material, at such a speed or exposure time as to completely

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burn out all the diazo and then develop. For format 3 film expose two sheets. An equivalent film area for other formats is to be used. Heat the exposed and developed strips in a forced air circulating oven for five minutes at 122° ± 2°F. to allow retained ammonia vapor to escape. Using a densitometer conforming to 4.3.2 check the zero and sensitivity settings of the densitometer with the appropriate working standards. For format 1 and 2 film take five density readings on the image side of the strips at two inch intervals. For format 3 film take five readings on the image side of each sheet, one in the center and one on each diagonal half-way to the edge. Determine the average of the readings taken and report the density as minimum density.

4.3.11 Actinic transmittance. On a commercial diazotype film processor, recommended by the film manufacturer, accurately calibrated to read linear speed and without the use of a master, expose and develop a sample of film under test at such a speed as to leave a density of 0.05 ± 0.01 above minimum (base) density. Record the true linear print speed as the "Maximum print speed of the reprint film." Without the use of a master, completely expose and develop a strip of the film under test, leaving a clean background. Using this film as the master, expose and develop a strip of the reprint film at such a speed as to leave a background density equal to that obtained with no master (0.05 ± 0.01 density.) This is the "Reprint speed of film." Determine the actinic transmittance by substituting the test value in the following equation and solving:

$$\text{Percent Actinic Transmittance} = \frac{\text{Reprint Speed of film}}{\text{Maximum print speed of reprint film}} \times 100$$

As an alternate method, a commercial diazotype film processor with an exposure shutter can be used. Repeatability of exposure times must be within ± 4 percent. The same procedure as above is followed, using this equation:

$$\text{Percent Actinic Transmittance} = \frac{\text{Maximum Exposure Time of Reprint Film} \times 100}{\text{Reprint Exposure Time of Film}}$$

4.3.12 Resolving power. A silver resolving test target (Contrast at least 100:1) as manufactured by Buckbee-Mears Company, St. Paul, Minn., or equivalent, is contact printed onto the test film in a vacuum frame. A mercury vapor lamp or a lamp of equal characteristics shall be used for exposure at a distance of 36 inches minimum. Exposure time shall be such that a barely visible density (0.05 above base) is achieved in the burnout area upon development of the image. Development shall be made on the developing section of a commercial ammonia processor recommended by the film manufacturer at such a speed as to achieve full image density. The film should be examined using an 80- to 100- power microscope, and the last group in which it is possible to count individual lines in both directions shall be the resolving power.

4.3.13 Accelerated aging.

4.3.13.1 Change of minimum density of light aged, processed film samples. Two samples, not less than six inches each for format 1 and 2 film nor less than 35mm x 148mm for format 3 (an equivalent film area for other formats is to be used) shall be cut from the fully exposed samples used in 4.3.10 and exposed in the Fade-O-Meter Model 18 as manufactured by the Atlas Electrical Devices Co., Chicago, Illinois or equal, for 3 hours at 160°F. black panel temperature with the humidifier switch off. After this exposure the samples shall be conditioned for three hours at 70°F. and 50 percent relative humidity. Take five density readings on the minimum image density area of each sample and determine the average reading. The difference between the figure obtained and the minimum image density as determined in 4.3 10 shall be recorded as the change of minimum image density of light aged samples.

4.3.13.2 Change of maximum density of light aged, processed film samples. Two samples, not less than six inches each for format 1 and 2 film nor less than 35mm x 148mm for format 3(an equivalent film area for other formats is to be used) film shall be cut from the developed samples used in 4.3.8. These are to be exposed in the Fade-O-Meter Model 18 as manufactured by the Atlas Electrical Devices Co., Chicago, Illinois or equal, for 3 hours at 160°F. black panel temperature with the humidifier switch off. After this exposure, the samples shall be conditioned for three hours at 70°F., and 50 percent relative humidity. Take five density readings on the maximum density area of each sample and determine the average reading. The difference between the figure obtained and the maximum image density as determined in 4.3.8 shall be recorded as the change of maximum density of light aged samples. Determine the percentage change by dividing the original desnity obtained in 4.3.8 into the density change, and multiplying by 100.

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4.3.13.3 Change of minimum density of heat aged, unprocessed film samples. An unopened production package of not less than 100 feet for format 1 and 2 film, nor containing less than 25 sheets of format 3 (an equivalent film area for other formats is to be used) film shall be placed in a 122° ± 2°F oven for a period of 72 hours. At the end of the test period, the package is removed from the oven and allowed to come to room temperature. For format 1 and 2 film select two strips at least 12 inches in length from the center of the roll, and for format 3 film select two sheets from the center of the package. Cover one-half of each sample with an opaque material and expose at a speed that will burn out all the diazo in the uncovered half of the sample. Completely develop the samples and heat them in a forced air circulating oven for five minutes at 122° ± 2°F. Take five density readings on the minimum image density area of each sample and determine the average reading. The difference between the figure obtained and the minimum image density as determined in 4.3.10 shall be recorded as the change of minimum image density of forced aged samples.

4.3.13.4 Change of maximum density of heat aged, unprocessed film samples. Take five density readings on the maximum density areas of the two test samples prepared in 4.3.13.3 and obtain the average value. The difference between the figure obtained and the maximum density as determined in 4.3.8 shall be recorded as the change in maximum density of forced aged samples. Determine the percentage change by dividing the original density obtained in 4.3.8 into the density change, and multiplying by 100.

4.3.13.5 Change of minimum density of heat aged, processed film samples. For format 1 and 2 film select two samples, 12 inches long and for format 3 film, two sheets (an equivalent film area for other formats is to be used.) Cover one-half of each sample with an opaque material and expose them on a diazo processor at a speed which will completely burn out all the diazo in the uncovered half of the samples. Develop the samples and then place them, sensitized side up, in a constant temperature oven at 122° ± 2°F, for a period of 72 hours. At the end of this time, remove the strips and take five density readings on the minimum density area of each sample. Obtain the average value. The difference between the figure obtained and the minimum image density as obtained in 4.3.10 shall be recorded as the change of minimum density of aged prints.

4.3.13.6 Change of maximum density of heat aged, processed film samples. Take five density readings on the maximum density areas of the two test samples prepared in 4.3.13.5 and obtain the average value. The difference between this figure and the maximum density as determined in 4.3.8 shall be recorded as the change in maximum density of aged prints. Determine the percentage change by dividing the original density obtained in 4.3.8 into the density change, and multiplying by 100.

4.3.14 Developing. All tests requiring development with or without an exposure step, shall be made in the developing section of a commercial ammonia printer processor which has been recommended by the film manufacturer.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packaging. The film shall be packaged in a close-fitting water and light resistant container. The container shall be sealed by means of tape conforming to PPP-T-60, type III or IV. The film shall be further packaged in quantities specified (see 6.2), in close-fitting boxes conforming to PPP-B-636, class domestic, style optional. Closure of boxes shall be in accordance with method II of the appendix to PPP-B-636.

5.1.1.2 Intermediate package. The film unit packaged, as specified in 5.1.1.1, shall be intermediate packaged in quantities specified (see 6.2), in close-fitting boxes conforming to PPP-B-636, class domestic. Closure of boxes shall be in accordance with method II of the appendix to PPP-B-636.

5.1.2 Level C. The film shall be packaged in accordance with the manufacturers standard practice, providing that this will insure protection for the film during shipment and provides for safe delivery to its destination.

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5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. The film packaged, as specified in 5.1, shall be packed in quantities, as specified (see 6.2), in a close-fitting box conforming to PPP-B-636, class weather resistant, grade V2s; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-640, class 2. The gross weight of the PPP-B-636 boxes shall not exceed 90 pounds and the gross weight of the other boxes shall be limited to 200 pounds. The wood boxes shall be provided with a waterproof case liner conforming to MIL-L-10547. Closure of boxes and case liners when applicable, shall be in accordance with the appendix to the applicable specification.

5.2.2 Level B. Unless otherwise specified, the film packaged, as specified in 5.1, shall be packed in quantities, as specified (see 6.2), in a close-fitting box conforming to PPP-B-636, class domestic. The gross weight of the boxes shall not exceed 90 pounds. The boxes shall be closed in accordance with method II of the appendix to PPP-B-636. When specified, (see 6.2), the boxes shall conform to PPP-B-636, class weather resistant, grade V3c or V3s.

5.2.3 Level C. The film packaged, as specified in 5.1, shall be packed in containers that will assure carrier acceptance and safe arrival at destination in compliance with the Uniform Freight Classification Rules and the National Motor Freight Classification Rules.

5.3 Palletization. When specified (see 6.2), the film packaged, as specified in 5.1, and packed, as specified in 5.2, shall be palletized as specified in MIL-STD-147.

5.4 Marking.

5.4.1 Unit package marking. Unless otherwise specified (see 6.2), commercial labeling will be acceptable provided the label includes in legible characters the following information: The manufacturers name, trade name of product, classification designation, Federal Stock Number, size (width and length) of sheet and roll film, number of sheets, expiration date and coating number. The label shall be attached in such a manner that it need not be damaged when the package is opened. When specified (see 6.2), the unit packages shall be marked in accordance with 5.4.2.1 or 5.4.2.2, as specified (see 6.2).

5.4.2 Intermediate package and shipping container marking. In addition, to the special marking required by 5.4.3, all marking shall be in accordance with 5.4.2.1 or 5.4.2.2, as specified (see 6.2).

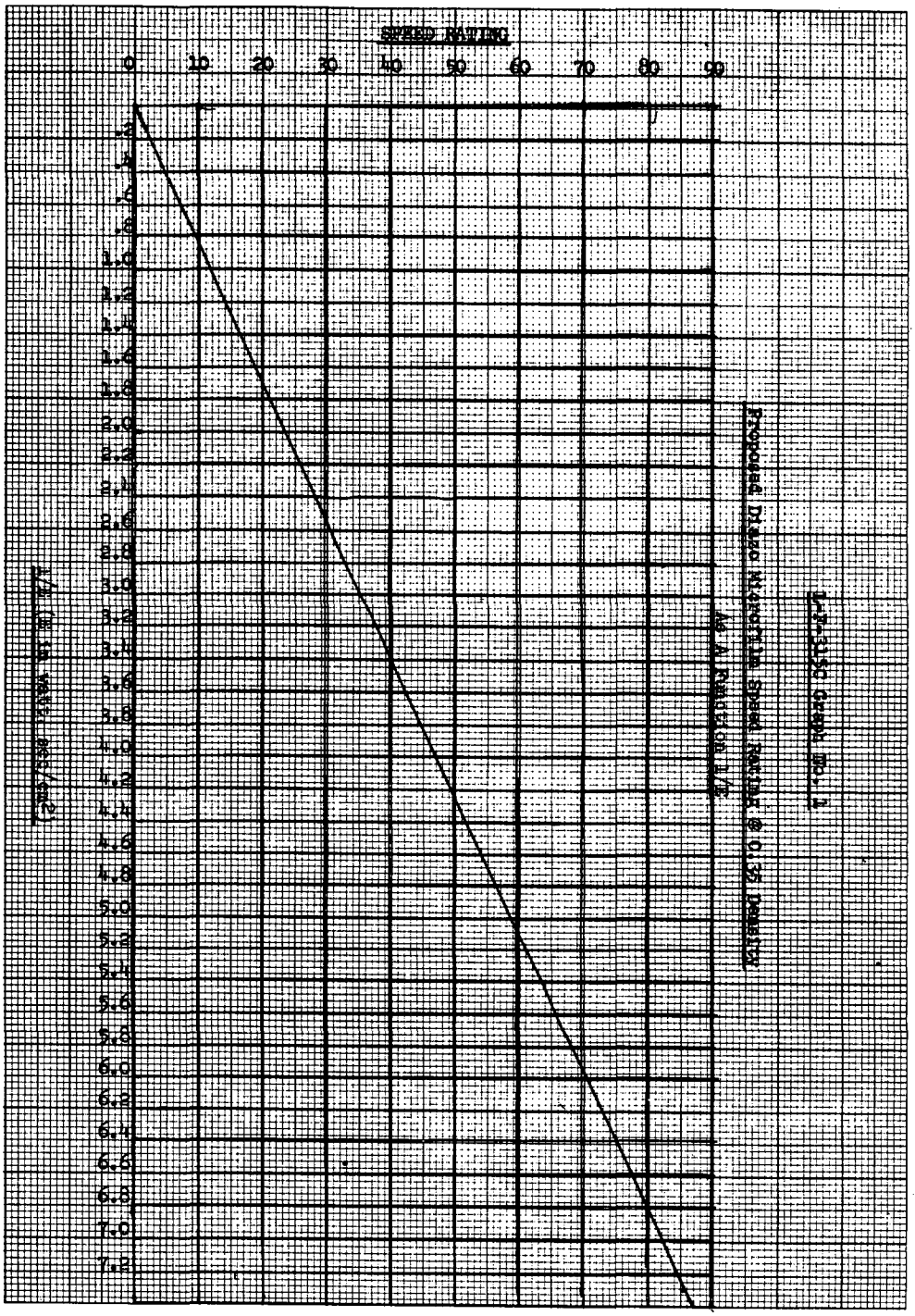
5.4.2.1 Civil agencies. In addition to the special marking required by the contract or order all intermediate packages and shipping containers shall be marked in accordance with Fed. Std. No. 123, and shall include the expiration date, classification designation and size.

5.4.2.2 Military agencies. In addition to the special marking required by the contract or order, all intermediate packages and shipping containers shall be marked in accordance with MIL-STD-129, and shall include the expiration date, classification designation and size.

5.4.3 Special marking. In addition to the markings required by 5.4.1 or 5.4.2, the following special marking shall be on all containers:

FILM, DIAZOTYPE, STORE IN A COOL DRY LOCATION
AWAY FROM AMMONIA - AVOID EXCESSIVE EXPOSURE TO
DAYLIGHT OR WHITE FLOURESCENT LIGHT.

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6 NOTES

6.1 Intended Use. The films covered by this specification are intended for use in microfilm equipment for the reproduction of negative or positive photographic images.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.

(b) Stock number.

(c) Classification designator (see table I).

(d) Rolls; perforated or unperforated length and width, wound coated side in or out and the type of spool or core (see table II).

(e) Sheets; size and quantity of sheet per package (see table II).

(f) If palletization is required so specify (see 5.3).

(g) Selection of samples (see 4.2 and table III).

(h) Level of packaging and packing (see 5.1 and 5.2).

(I) Exception to expiration date (see 3.3).

(j) Marking, if other than specified (see 5.4).

6.3 Definitions:

6.3.1 ARMM: Association of Reproduction Materials Manufacturer's, Inc.

6.3.1.1 ARMM sensitometer. A sensitometer designed and developed for the reproduction industry by the Association of Reproduction Materials Manufacturer's, for the precise and reproducible measurement of the light sensitivity of diazo materials.

6.3.2 Contract data requirements. Data conforming to Data Item Description DI-E-1295 will usually be required for delivery in connection with this specification. When so required, such data will be specified for delivery on a DD Form 1423, included in the contract.

MILITARY CUSTODIANS:

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Navy - AS
Air Force - 70

Preparing activity:

Navy - AS

Review activities:

Army - EL
Navy - AS, OS, MC
Air Force -70

CIVIL AGENCY INTEREST:

GSA-FSS

User activities:

Army - EL
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